Architectural Specification and Domain Compendium of the Reverse-the-Gap Decompilation Framework

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Level 1 System Context and the Reverse-the-Gap Decompilation Paradigm

Architectural Inversion: Pre-Compiled Graded Substrates vs. Accumulative Execution

Conventional theoretical computer science and mathematical physics model computation as a forward, accumulative process in which axiomatic primitives are posited, state transitions are executed sequentially, and terminal scalar readouts are treated as the primary ontology. When a forward model encounters an underdetermined transition, an apparent loss of entropy, or a non-invertible compression boundary, classical analysis attributes the incompleteness to the underlying object. The Reverse-the-gap architecture synthesized across the NEXUS corpus establishes the exact inverse software paradigm: the transformation domain is complete at step zero. Every finite word of every length, every coordinatewise displacement, and every lawful transition already exists within a single, label-free graded substrate $S[b, n]$. Computation neither creates nor destroys information; rather, it is a deterministic geometric folding and indexing process in which structure is rendered exclusively by reading what is already present.

Across all historical and active modules of the research programme, the entire software architecture is governed by one central specification question:

What distinctions does a transformation system make possible, and which of those distinctions can survive and be read?

To formalize this question as executable software documentation, the framework models any computational or physical domain as a typed four-tuple (X, T, Y, R), where X is the underlying state space, $T: X \to X$ is the state transformation, Y is the readout space, and $R: X \to Y$ is the reader or observation map.

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Standard forward engineering conflates the properties of the state transition T with the properties of the observation map R. The @reverse-the-gap decompilation protocol strictly separates three orthogonal architectural invariants: first, whether the state transformation T is bijective and thus invertible; second, the cardinality and structure of the observation fibers $R^{-1}(y)$ for any readout $y \in Y$; and third, which route histories and coordinate orderings are preserved or discarded by the representation.

An invertible state transformation T frequently feeds a many-to-one readout R, and a many-to-one readout can still possess uniform, algebraically countable fibers. Instead of stepping forward toward an unknown state, a decompiler begins with the finished observation $y \in Y$, subtracts the forward transport route, and binds the surviving algebraic residual to the specific cut and transformation that own it. Within this architecture, a "gap" is never a missing piece of reality; it is the exact mathematical signature of a reader's finite cut, projection fiber, or declared equivalence relation. While a collapsed boundary reading evaluates to a binary extreme (0 or 1), "potential"—formalized as the uncollapsed 0.5 midpoint between binary states—is the active space of admissible continuations held within a configuration fiber prior to scalar projection.

Dependency-Injection Container Semantics and Operational Primitives

To prevent the exponential state-space explosion characteristic of brute-force candidate generation, the reverse-the-gap engine operates as a strict Dependency-Injection (DI) container. In software engineering, dependency injection decouples the declaration of a required interface from the concrete provider that satisfies it. Applied to algebraic decompilation, the system never guesses values to fill an unknown state; instead, it subtracts every dependency already accounted for by the forward construction and inspects the exact algebraic remainder. In coupled discrete systems, this remainder is never random noise—it is an earlier structural layer re-addressed and read by an opposing wave.

The operational semantics of the DI container are governed by five foundational routing primitives and three variable-binding states, which standardize how state transformations and readouts are documented across all domains.

Primitive / Binding State	Architectural Category	Formal Structural Definition	Computational & Verification Semantics
WHAT	Write Primitive	A state transformation $F: X \to X$ that intrinsically carries its own algebraic inverse F^{-1}.	Enforces the invariant that no valid state write exists without a deterministic "way home" ($F^{-1}(F(x)) = x$).
WHERE	Read Primitive	The identical underlying structure interrogated at a spatial, angular, or temporal offset by a reader R.	Separates the label-free binary substrate from the equivalence vocabulary of the probing interface; writes and reads are never concurrent.

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Primitive / Binding State	Architectural Category	Formal Structural Definition	Computational & Verification Semantics
ROUTE	Traversal Mode	Directed navigation across a known coordinate offset between the WHAT and the WHERE.	Resolves target state addresses in $O(n)$ arithmetic operations proportional to register width n without materializing the state space.
SEARCH	Failure Mode	Unstructured enumeration across a discarded or unknown coordinate offset.	Scales with domain cardinality $O(2^n)$; represents the operational failure mode of unindexed reading.
HOLE	Analytical Unit	A sealed WHAT committed to the system prior to the resolution of its corresponding WHERE.	Defines the exact boundary constraints and neighbor relations that the DI container must bind without candidate scanning.
BOUND	DI Container State	A single algebraic provider uniquely determines the target variable across the interface.	The DI container resolves the coordinate or bit assignment deterministically with zero branching.
AMBIGUOUS	DI Container State	Multiple candidate providers satisfy the local constraint equation simultaneously.	Requires injecting an independent orthogonal readout (such as a retained high bit or middle value bit) as a qualifying selector.
UNBOUND	DI Container State	No active provider reaches the target variable under the current elimination rules.	Signals a topological obstruction (such as reconvergent fanout) and names the exact algebraic link to be constructed next.

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Formal Verification Ledger Protocol and Provenance Discipline

Unlike speculative theoretical frameworks that conflate mathematical theorems, numerical coincidences, and physical metaphors, the @reverse-the-gap software specification enforces a strict, append-only verification ledger. Every mathematical statement, algorithmic route, and cross-domain mapping is assigned an explicit status tag. The word CONFIRMED is permanently banned from the specification vocabulary to prevent heuristic confirmation bias. Instead, claims are governed by cryptographically sealed predictive execution: a formal prediction text and its verification script are written and hashed via SHA-256 (SEALED) before the test is executed, and the runner aborts immediately if the byte-level digest mismatches.

Ledger Status	**Operational Definition in the Decompilation Protocol**	**Canonical Instantiation in the NEXUS Corpus**
STANDS	Established theorem, proved algebraic identity, or externally supplied empirical dataset cited as an anchor.	Burnside–Möbius necklace counts, Boolean Möbius inversion, Lucas Fermat prime splitting, Schwinger spin construction, Landauer erasure floor ($k_B T \ln 2$).
RUNS	Computationally certified law where the formal route generated the exact landing prior to or independently of data linkage.	Width-3 Toffoli exchange (52eca8c4…), 64-bit π ring address ($2{,}427{,}008{,}858{,}123{,}944{,}441$), ROBDD 1-byte SHA-256 preimage collapse, atomic closures ($2,10,18,36,54,86,118$).
SEALED	Prediction text and audit source code hashed with SHA-256 prior to execution and verified unchanged post-execution.	All-word rank identity (5cde4126…), comparator imbalance spectrum (21895d2d…), Walsh parity split (be9c1901…), pullback duality (359b4d3e…).
DEAD END	A forward hypothesis or shortcut formula that executed cleanly but failed validation, subsequently routed to its true structural owner.	Literal half-width palindrome order copy (owned by binary counting order), smooth even-body closure (owned by address digit sum), SHA route-code inversion (owned by codec relabeling).
OPEN	A sealed boundary question where the formal intertwining map, closed-form spectrum, or	Regularity of the radial profile language (≥ 312 states), closed-form Walsh spectrum of latitude in

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Ledger Status	Operational Definition in the Decompilation Protocol	Canonical Instantiation in the NEXUS Corpus
	selector has not yet been written.	ring coordinates, wide-interface ROBDD region pruning.
PULL	An interpretive bridge connecting an exact finite compiler invariant to a broader physical, biological, or cosmological thesis.	Interpreting combinatorial sphere latitude walks as physical spin-$\frac{1}{2}$ dynamics, equatorial folds as Compton rest mass ($\omega = mc^2/\hbar$), and DNA as a 33 Hz IFFT bitstream.

Level 2 Container Architecture and Cross-Resolution Interfaces

The Six Core Execution Containers of the NEXUS Compiler Space

Following the C4 software architecture standard—which decomposes a system from high-level Context (Level 1) into independently deployable Containers (Level 2), functional Components (Level 3), and Code/Formula implementations (Level 4)—the NEXUS transformation universe resolves into a single operator-and-access program distributed across six coupled containers:

C0: Distinction → C1: Lawful Continuation → Transformation Domain → Configuration / Relation Space → Quotient &Residual → Reader / Access Operator → Observable Distinction

Each previously separate research subject in the corpus operates as a specialized test environment (Container) verifying a specific stage of this pipeline.

Container ID	Container Name	Primary Mathematical Arena	Core Responsibility & Exported Interface	Primary Reference Modules
CNT-01	**Graded Compiler Kernel**	$S[b,n] = \{0,\dots,b-1\}^n$ and $\mathrm{GF}(2)^n$ [cite: 1, 9]	Manages the complete word inventory, uniform difference fibration Δ, recursive operator-table	nexus_compiler_space.py, nexus_reverse_gap_width3.py [cite: 1, 10]

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Container ID	Container Name	Primary Mathematical Arena	Core Responsibility & Exported Interface	Primary Reference Modules
			wrapping $L \cdot b^L$, and width-3 Toffoli instruction closure.	
CNT-02	**Addressable Binary Ring Engine**	Circular ordering R_n of 2^n words under P_n [cite: 4, 7]	Implements the fixed-point-free partner involution P_n, $O(n)$ closed-form prefix ranking via comparator imbalance $\Delta(p)$, and the $L \to L+1 \to L+2$ mirror lift codec.	rule_based_ring.py, paper_checks.py, pullback_run.py [cite: 4, 7, 16]
CNT-03	**Spherical & Spectral Waveform Engine**	Discrete sphere (r, z, θ) and Walsh basis H_{2^n} [cite: 7, 11]	Executes abelianized Hamming-weight projections, Walsh-even/odd parity splitting, isometric half-width carrier transport T_n, and seam/closure	walsh_split.py, seam_closure.py, trade_measure.py [cite: 7, 8, 16]

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Container ID	Container Name	Primary Mathematical Arena	Core Responsibility & Exported Interface	Primary Reference Modules
			digit-sum readouts.	
CNT-04	**Circular Tape & Radix Lane Transducer**	Cyclic windows $W_{L,i}$, Galois core R_0, BBP lanes	Couples radial compilation with angular execution on the 2-bit ring, conjugates circular shifts into Boolean Möbius differences ($MR = TM$), and resolves Galois-stable zero routes.	two_bit_circle.xlsx, DI_chain_rollup.ipynb, Nest.docx [cite: 1, 4, 5]
CNT-05	**Homological Cut & Carry-Capacity Analyzer**	Cubical complexes $C_G^{\bullet}(X, K)$ and Myhill–Nerode cuts	Evaluates persistence residuals $\varepsilon_{\text{harm}}$ across token arenas, computes disjoint-square adder cohomology $b_1^{\text{disj}}(n)$, and enforces constant-transfer cut separability $b^g \geq m$.	rings.py, fold.py, seam.py [cite: 1, 3, 6]

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Container ID	Container Name	Primary Mathematical Arena	Core Responsibility & Exported Interface	Primary Reference Modules
CNT-06	**Cryptographic Cofactor & Substrate Router**	Boolean DAGs $\Psi_H(\mu)$, ROBDDs, and 2ℓ-bit shells	Compiles 64-round SHA-256 inverses into canonical ROBDDs via Shannon cofactoring, bijects 512-bit schedule tails $E: B \leftrightarrow T$, and projects $2n^2$ atomic/nuclear shell closures.	shared_carry_inverse.py, cofactor_inverse_bdd.py, nuclear_unfold.py [cite: 1, 6]

Cross-Resolution Interface Honesty and Functorial Commutation

When connecting two containers operating at different levels of resolution—such as mapping a continuous physical double-well potential to a binary bit, a bit vector to a logical register, or a register to an arithmetic adder—the software specification requires a formal audit of interface integrity. Let $(C^\bullet)^G \overset{\iota}{\hookrightarrow} C^\bullet$ denote the inclusion of the restricted interface complex into the full object complex, and let $\iota_*: H^q((C^\bullet)^G) \to H^q(C^\bullet)$ (or the projection Π inducing $H^q(\Pi)$) denote the induced map on cohomology. Every cross-layer interface decomposes into three exact cohomological sectors:

$\mathrm{im}\iota_*$:residual structure genuinely carried through the interface

$\ker\iota_*$:restricted-interface cohomology made trivial by the full object (reader-only artifacts)

$\mathrm{coker}\iota_*$:full-object cohomology inaccessible to the chosen interface (hidden object residuals)

For a chosen target set of distinctions $\mathcal{T}$ (or linear subspace T), an interface is certified as **honest** if and only if it preserves the continuation class required by the next layer, introduces no spurious reader-only residual in the target sector, and hides no required object-level residual:

Honest Interface
$\Leftrightarrow$ Commuting Implementation Square $+$ $((\mathcal{T} - \mathcal{T}) \cap \ker\iota_* = \{0\})$
$+$ Target-Safe Cokernel

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In an end-to-end physical-to-logical trace, this cohomological diagnostic pinpoints exactly where naive abstractions fail across three canonical engineering rungs. At the first rung, mapping a continuous double-well coordinate x to a single bit via a basin decoder $\pi(x) \in \{0,1\}$ places the interface boundary at the phase-space separatrix; interface failure manifests as metastability and finite-resolution dwell-time ambiguity when probability mass lingers near the 0.5 separatrix. At the second rung, lifting individual bits into a multi-bit register treats independent single-bit flips as commuting disjoint-support squares ($F_i F_j - F_j F_i = 0$), where physical cross-talk or correlated error channels surface as non-zero square defects. At the third rung, coupling registers into a binary adder introduces non-local carry propagation that violates disjoint-support independence, concentrating a non-zero commutation residual $\Delta_{\text{carry}} = F_i F_j - F_j F_i \neq 0$ precisely along the carry boundaries.

Level 3 Component Specifications and Domain Formula Compendium

Component 3.1: Graded Kernel, Difference Fibration, and Width-Three Instruction Closure

The Graded State Space and Difference Fibration

At the foundation of CNT-01 (nexus_compiler_space.py), the state layer for any integer base $b \geq 2$ and word width $n \geq 0$ is the finite Cartesian power $S[b,n] = \{0,1,\ldots,b-1\}^n$ with cardinality $|S[b,n]| = b^n$. The coordinatewise displacement (difference) operator $\Delta: S[b,n] \times S[b,n] \to S[b,n]$ is defined by:

$$\Delta(x,y)_i = (y_i - x_i) \bmod b \quad \forall i \in \{0,\ldots,n-1\}$$

Verified across 22 base and length configurations ($b = 2, n = 0 \ldots 8$; $b = 3, n = 0 \ldots 5$; $b = 4, n = 0 \ldots 4$) and 89,405 recursive edge checks, Δ obeys three structural laws that govern the substrate. First, by surjective layer identity, the image $\mathrm{Im}(\Delta)$ is the entire word layer $S[b,n]$ itself, proving that relational displacements inhabit the exact same inventory as states rather than an external metadata space. Second, by uniform preimage fibration, for every displacement word $\delta \in S[b,n]$, the preimage fiber $\Delta^{-1}(\delta) = \{(x,y) \in S[b,n]^2 : y \equiv x + \delta \pmod b\}$ has uniform cardinality $|\Delta^{-1}(\delta)| = b^n$, because $y = (x+\delta) \bmod b$ is uniquely determined for each of the b^n choices of x. Calling the codomain the same word space is mathematically exact, whereas calling Δ an isomorphism is a type error that erases its many-to-one fibration of b^{2n} ordered pairs onto b^n displacements. Third, by edge-split recursion, appending a new digit splits the difference cleanly at the register boundary according to $\Delta(x \parallel a, y \parallel c) = \Delta(x,y) \parallel ((c-a) \bmod b)$.

Recursive Wrapper Encoding and Operator-Space Reflection

Any total transformation $T: S[b,n] \to S[b,m]$ is completely specified by concatenating its b^n output words of width m in canonical lexicographic input order. Consequently, the operator space $\mathcal{T}[b, n \to m]$ has cardinality $b^{m \cdot b^n}$, and every operator is natively encoded as a single state word of width $L_{\text{table}} = m \cdot b^n$ inside the higher layer $S[b, m \cdot b^n]$. For endomorphisms ($n = m = L$), iterating the wrapper map $L_{d+1} = L_d \cdot b^{L_d}$ from $L_0 = 1, b = 2$ generates an unbounded tower of finite compiler descriptions:

$$L_0 = 1 \;\to\; L_1 = 1 \cdot 2^1 = 2 \text{ (4 maps)} \;\to\; L_2 = 2 \cdot 2^2 = 8 \text{ (256 maps)} \;\to\; L_3 = 8 \cdot 2^8 = 2{,}048 \text{ } (2^{2048} \text{ maps}) \;\to\; L_4 = 2048 \cdot 2^{2048}$$

The sequence of endomorphism table lengths $L \cdot 2^L$ (2,8,24,64,160,384,896,2048, …,49152 at $L = 12$) is identical to the total number of angular bit slices $n \cdot 2^n$ in the nested polar rings. At $L = 2$, the truth-table

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width $2 \cdot 2^2 = 8$ bits matches the 8-slice circumference of the 2-bit ring, establishing a canonical bijection between the 256 operators $\mathcal{T}[2,2 \to 2]$ and the 256 byte states of $S[2,8]$.

The Width-Three Exchange Theorem and Instruction Set Closure

A core formal proof of the @reverse-the-gap compiler (nexus_reverse_gap_width3.py, sealed under SHA-256 52eca8c48a887f4fb9235a1a4c297d701a9b83f8af0180a93c8cb25ac6b4506e) derives the minimum binary register width n required to support a transformation $F: \mathrm{GF}(2)^n \to \mathrm{GF}(2)^n$ that simultaneously satisfies three architectural constraints: Reversibility (F is a bijection with an exact return route F^{-1}), Fixed-Point-Freeness ($F(x) \neq x \; \forall x \in \mathrm{GF}(2)^n$, meaning no state can hold still), and Nonlinear Interaction (at least one coordinate of F contains an Algebraic Normal Form (ANF) monomial of degree ≥ 2).

Over $\mathrm{GF}(2)^n$, the affine general linear group $\mathrm{AGL}(n, 2)$ consists of all invertible maps of the form $x \mapsto Ax \oplus v$ with $A \in \mathrm{GL}(n, 2)$, which all have ANF degree ≤ 1. Its exact cardinality is:

$$|\mathrm{AGL}(n, 2)| = 2^n |\mathrm{GL}(n, 2)| = 2^n \prod_{i=0}^{n-1} (2^n - 2^i)$$

Comparing $|\mathrm{AGL}(n, 2)|$ against the symmetric group $|\mathrm{Sym}(2^n)| = (2^n)!$ across register widths proves why nonlinear reversible computation is impossible below width three. At width $n = 1$, $|\mathrm{AGL}(1,2)| = 2^1(2^1 - 1) = 2 = 2! = |\mathrm{Sym}(2)|$, so every bijection is affine; the 4 total maps partition by image rank into 2 bijections of rank 2 (pass $x \mapsto x$ and flip $x \mapsto x \oplus 1$) and 2 constant merges of rank 1. At width $n = 2$, $|\mathrm{AGL}(2,2)| = 2^2(4-1)(4-2) = 24 = 4! = |\mathrm{Sym}(4)|$, proving that every 2-bit bijection is likewise affine and zero nonlinear permutations exist. Across all $2^8 = 256$ endomorphisms on $S[2,2]$, the image-rank distribution is $\{1: 4,\ 2: 84,\ 3: 144,\ 4: 24\}$, where spanning the full landscape requires only two new primitives beyond 1-bit gates and wire swaps: the asymmetric reversible CNOT mixer $(a, b) \mapsto (a, a \oplus b)$ and the irreversible rank-3 AND merge $(a, b) \mapsto a \wedge b$. At width $n = 3$, however, $|\mathrm{AGL}(3,2)| = 2^3(8-1)(8-2)(8-4) = 8 \times 168 = 1{,}344 < 8! = 40{,}320$, opening a gap of 38,976 non-affine reversible permutations.

Rather than scanning the $40{,}320$ permutations, the compiler constructs the explicit width-three provider by conjugating an affine fixed-point-free involution X with the self-inverse nonlinear Toffoli gate T:

$$T(a, b, c) = (a,\ b,\ c \oplus ab), \qquad X(a, b, c) = (a \oplus c,\ b \oplus 1,\ c), \qquad F = T \circ X \circ T$$

Because $T^2 = I$ and $X^2 = I$ with $\mathrm{Fix}(X) = \varnothing$, the conjugate map F automatically satisfies $F^2 = TXT^2XT = I$ (so $F^{-1} = F$ is its own way home) and $\mathrm{Fix}(F) = T(\mathrm{Fix}(X)) = \varnothing$. Evaluating $F(a, b, c) = T(a \oplus c,\ b \oplus 1,\ c \oplus ab)$ yields the exact compiled ANF coordinates via the Boolean Möbius transform $c_S = \bigoplus_{U \subseteq S} f(U)$:

$$a' = a \oplus ab \oplus c \quad (\deg 2), \qquad b' = 1 \oplus b \quad (\deg 1), \qquad c' = a \oplus bc \quad (\deg 2)$$

A critical algebraic distinction recorded in the ledger is the **Degree Correction**: for any bijection on $\mathrm{GF}(2)^n$ ($n \geq 2$), balancedness forces the top-degree n ANF monomial (abc at $n = 3$) to vanish identically in every coordinate. Thus, the surviving nonlinear terms at width 3 have degree $n - 1 = 2$ (quadratic), not degree 3. Above width 3 ($n \geq 4$), gates of width $\leq n - 1$ generate the alternating group A_{2^n}, and borrowing one auxiliary lane recovers all odd permutations; similarly, any invertible n-bit feedback shift register

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$F(x_0, \ldots, x_{n-1}) = (x_1, \ldots, x_{n-1}, x_0 \oplus g(x_1, \ldots, x_{n-1}))$ ($2^{2^{n-1}}$ choices) embeds an AND-type merge inside g. Consequently, the fundamental logical instruction set closes permanently at width three.

Component 3.2: Addressable Binary Rings, Base-b Admissibility, and the Mirror Lift Codec

Partner Involution and the Even-Base Admissibility Theorem

Container CNT-02 constructs a complete, duplicate-free circular address system R_L for all b^L words of length L over an alphabet of base b. The engine's defining operator is the conditional partner map $P_b: S[b, L] \to S[b, L]$, which reverses the digits of a word unless the word is a palindrome, in which case it complements each digit via $d \mapsto b - 1 - d$:

$$P_b(x) = \begin{cases} \mathrm{rev}(x), & x \neq \mathrm{rev}(x), \\ \mathrm{comp}_b(x), & x = \mathrm{rev}(x). \end{cases}$$

Extending the ring investigation beyond binary to arbitrary bases b establishes the Even-Base Admissibility Theorem (SEALED under a9a396ec…), which proves that P_b is a fixed-point-free involution ($P_b^2 = I$ and $\mathrm{Fix}(P_b) = \varnothing$) if and only if the base b is even. For any non-palindrome, digit reversal satisfies $\mathrm{rev}(x) \neq x$ and $\mathrm{rev}(\mathrm{rev}(x)) = x$. For a palindrome, $\mathrm{comp}_b(x)$ remains a palindrome, and a fixed point occurs if and only if every digit satisfies $d = b - 1 - d \Leftrightarrow 2d = b - 1$. An integer digit $d \in \{0, \ldots, b-1\}$ satisfies $2d = b - 1$ if and only if b is odd, in which case the constant middle-digit word $\left(\frac{b-1}{2}\right)^L$ is fixed by P_b, violating the C1 continuation law ("no word holds its place"). Binary ($b = 2$) is therefore not an arbitrary engineering choice, but the minimal admissible base.

In general even bases, the digitwise residual $Q_b(x) = x - P_b(x) \pmod b$ separates into two distinct reflection symmetries: for a non-palindrome ($P_b(x) = \mathrm{rev}(x)$), $Q_i = x_i - x_{L+1-i} \pmod b$ is **anti-palindromic** ($\mathrm{rev}(Q_b(x)) = -Q_b(x) \pmod b$), whereas for a palindrome ($P_b(x) = \mathrm{comp}_b(x)$), $Q_i = 2x_i - (b-1) \pmod b$ is **palindromic** ($\mathrm{rev}(Q_b(x)) = Q_b(x)$). Base $b = 2$ is the unique even base where $-q \equiv q \pmod 2$, collapsing both reflection classes into a single palindromic XOR residual $Q_2(x) = x \oplus P_2(x)$ (1^L on palindromes). When the binary ring R_L is flattened into a bitstring S_L of length $L \cdot 2^L$, reflecting across $q^\star = L \cdot 2^L - 1 - q$ yields the exact palindrome detector mask $S_L[q] \oplus S_L[q^\star] = 1$ if the word containing q is a palindrome and 0 otherwise.

The Folded Square, Base-Independent Parabola, and Continuation Attractors

Selecting the numerically smaller member of each partner pair as the canonical representative ($x < P_b(x)$, yielding $M = b^L/2$ representatives), the ring layout places the sorted representatives $a_0 < a_1 < \cdots < a_{M-1}$ in ascending order on the left face ($0 \ldots M-1$), places their partners in descending order on the right face ($M \ldots N-1$), and swaps the center seam pair at positions $M-1$ and M. This layout satisfies the universal facing law at every position:

$$R_L[b^L - 1 - i] = P_b(R_L[i]) \quad \Leftrightarrow \quad R_L \circ \sigma_L = P_b \circ R_L$$

To analyze the ring geometrically without enumerating it, split an even-width word $x = f \parallel g$ ($L = 2h, H = b^h$) into its front half f and back half g, and plot x in the unit square at $(\varphi, \gamma) = (f/b^h, \mathrm{rev}_h(g)/b^h)$. Peeling

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the outermost digit pair (d, e) defines a $b \times b$ decision grid: cells with $d < e$ are immediately canonical (YES), cells with $d > e$ are partners (NO), and the diagonal $d = e$ defers the comparison inward (MAYBE):

$$N_{\text{yes}} = \frac{b(b-1)}{2}, \qquad N_{\text{no}} = \frac{b(b-1)}{2}, \qquad N_{\text{maybe}} = b, \qquad \text{Maybe Fraction} = \frac{b}{b^2} = \frac{1}{b}$$

After $k = \lfloor L/2 \rfloor$ inward comparisons, the surviving diagonal words number $b^L b^{-\lfloor L/2 \rfloor} = b^{\lceil L/2 \rceil}$—the exact palindrome count. Because representatives occupy the upper triangle ($\gamma > \varphi$) plus the lower half of the diagonal ($\gamma = \varphi < 1/2$), the density of representatives at normalized value $t = X/b^L$ is the base-independent straight line $F'(t) = 1 - t$, and its integral is the universal angle parabola:

$$C_{b,L}(X) = X - \frac{1}{2} J_{b,L}(X) = C_L^{\text{rev}}(X) - \frac{1}{2} K_{b,L}(X) \quad \Rightarrow \quad F(t) = t - \frac{t^2}{2}$$

where $K_{b,L}(X) = \#\{w = \text{rev}(w) : N - X \le w < X\} \le b^{\lceil L/2 \rceil}$ isolates the exact diagonal palindrome tent $\frac{1}{2}\min(t, 1-t)$. For binary ($b = 2$), the number of representatives starting with a p-bit prefix q ($2p \le L$) has the exact finite block law $N_L(q) = 2^{L-p}(1 - \text{mid}(q)) + 2^{\lceil (L-2p)/2 \rceil}([q_0 = 0] - \frac{1}{2})$. Consequently, a radial reader interrogating angle $u < 1/2$ lands in a word whose front half directly encodes the binary digits of the square root $\tau(u) = 1 - \sqrt{1 - 2u}$ (verified on 3,891/3,892 readings). Under the four frame providers (00, 01, 10, 11), stripping frames inward defines the ancestry continuation operator $F_L(a) = \min(1a1, P_{L+2}(1a1))$: one provider in four continues ($\Pr(\text{depth} \ge t) = 4^{-t}$, mean depth → 1/3), the reversal branch $1a1$ contracts toward the seam $u = 1/2$ via $u \mapsto 3/8 + u/4$, and the palindrome complement branch $0C(a)0$ maps $\varphi \mapsto (1 - \varphi)/2$, converging to the fixed-point attractor $\varphi = 1/3$ at angle $u(1/3) = 1/3 - 1/18 = 5/18$ (partner $2/3$ at $13/18$).

Single-Bit Width Progression and the Mirror Lift Route Codec

While the pad law $F_L(a)$ steps across even or odd rings by two bits ($L \to L + 2$), closing the single-bit $L \to L + 1$ hole (SEALED under ab543c24… and 359b4d3e…) reveals that the growth route of a word is the Boolean Möbius butterfly applied across the mirror. In the even-to-odd step ($L = 2h \to L + 1$, **Theorem 3**), the middle bit of an odd word is compared only with itself, making it the first free bit in each block. Inserting $b \in \{0,1\}$ at the middle ($I_b(a)$) maps a canonical parent at descending offset $q = D_K + i$—where $K = H - 1 - A$, $D_K = \frac{K(K-1)}{2} + \max(0, K - H/2)$, and $S_K = K + [K \ge H/2]$—to the exact odd child offsets:

$$q_1 = 2D_K + i \quad \text{(middle 1)}, \qquad q_0 = 2D_K + S_K + i \quad \text{(middle 0)}$$

Both children share the identical pair residual $Q_{L+1}(I_b(a))$ (the parent residual with a middle 0, or all ones for palindromes), rendering the readout-to-address link AMBIGUOUS until qualified by the retained middle value bit m.

In the subsequent odd-to-even step ($L + 1 \to L + 2$, the **E/G Lift**), each odd representative with middle bit m generates two canonical representatives at width $L + 2$, thereby resolving the "homeless" middle-zero branch:

$$E_m(x) = \text{canon}_{L+2}(mxm), \qquad G_m(x) = 0P_L^m(x)1$$

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In this E/G gauge, the branch bit $d = x_0 \oplus x_{L+1}$ is the XOR residual of the new outer pair, and m is directly readable from the finished word as the outer frame bit when $d = 0$ (modulo the palindrome exception $E_1(x) = 0C(x)0$) or as the canonical face of the inner core when $d = 1$. Defining the mirror butterfly on pair $j \in \{0, \dots, h-1\}$ ($j = 0$ outermost) by value $v_j = x_j$ and difference $\delta_j = x_j \oplus x_{n-1-j}$, **Theorem 4** proves that growing any n-bit word from the 1-bit root 0 via odd middle-insertions (m) and even E/G steps (d), plus a final partner-side bit $\varepsilon = [x \neq \mathrm{canon}(x)]$, establishes an exact n-bit bijection in which the route's even-step residual bits are precisely the mirror differences $\delta_{h-1}, \delta_{h-2}, \dots, \delta_0$, recorded innermost first.

Component 3.3: Comparator Imbalance, Gate-Level Borrow Chains, and All-Word Prefix Ranking

The central theorem of *Against Its Own Reverse* (CNT-02 / CNT-03) demonstrates that the entire binary ring R_n, its $O(n)$ address function, and its spectral carrier are built from a single operation: comparing a binary word x against its own bit-reversal $J(x)$. Define the comparator sign:

$$c(x) = \mathrm{sgn}(J(x) - x) \in \{-1, 0, +1\}$$

A word x is a canonical representative if and only if $c(x) = +1$, or $c(x) = 0$ and $x_0 = 0$. At the hardware gate level (**Theorem 7.2**, SEALED under 5910642e…), $c(x)$ is the difference of the terminal borrows of two ripple subtractors:

$$c(x) = \beta(x - J(x)) - \beta(J(x) - x), \qquad \beta_{\text{out}} = \mathrm{Maj}(\neg a, b, \beta_{\text{in}})$$

Thus, the ring computes its own addresses using the exact full-adder carry gate (a 3-input majority gate) with one input complemented, applied across mirrored bit pairs. At width $n = 5$, the first nonlinear degree-mixing correction in the palindrome address $i_5(u)$ for a 3-bit half-word $u = 4u_2 + 2u_1 + u_0$ is literally a single majority borrow gate:

$$i_5(u) = 4u + 2u_2 + \mathrm{Maj}(1 - u_0, u_1, u_2)$$

Summing $c(x)$ across all integers below a value cut $p \in \{0, \dots, 2^n\}$ defines the **Comparator Prefix Imbalance** $\Delta(p)$ (**Proposition 2**):

$$\Delta(p) = \sum_{x<p} c\,(x) = \#\{x : x < p \leq J(x)\} \geq 0$$

Because every non-palindromic reversal pair $\{x, J(x)\}$ with $x < J(x)$ contributes $+1$ at x and -1 at $J(x)$, $\Delta(p)$ counts the exact number of reversal pairs separated by the cut at p—the relational carry that a reader standing at threshold p must hold across the cut.

Theorem 1 and Theorem 7.3: Closed-Form All-Word Address and Rectangle Digit Sum

For any threshold p, the number of canonical representatives strictly below p is given by the exact closed-form identity (**Theorem 1**, verified across all $2{,}097{,}148$ thresholds for $n = 2 \dots 20$ and scalar widths up to $1{,}024$):

$$\mathrm{rank}(p) = \frac{1}{2}(p - \pi(p) + \Delta(p)) + \pi_c(p)$$

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where $\pi(p) = u + [B(u) < p]$ counts all palindromes below p (read directly from the top $\lceil n/2 \rceil$ bits u of p and its palindrome reconstruction $B(u)$), and $\pi_c(p) = \min(\pi(p), 2^{\lceil n/2 \rceil - 1})$ counts the canonical palindromes below p. When p is a canonical palindrome with half-word u, $\pi(p) = \pi_c(p) = u$, simplifying to **Theorem 7.1**: $i_n(u) = \frac{1}{2}(p + u + \Delta(p))$. A canonical word a occupies physical ring address $\mathrm{rank}(a)$ (except the largest canonical word $\mathrm{rank}(a) = M - 1$, which the seam swap places at M), and its partner occupies $N - 1 - \mathrm{pos}(a)$.

Crucially, $\Delta(p)$ requires zero interval enumeration (**Theorem 7.3**, SEALED under 21895d2d…). With $h = \lfloor n/2 \rfloor$ and $H = 2^h$, an even-width threshold $p = AH + B$ ($n = 2h$) and an odd-width threshold $p = (2A + b)H + B$ ($n = 2h + 1$, middle bit $b \in \{0,1\}$) evaluate in closed form as:

$$\Delta_{2h}(p) = A(H - A) + B - 2\mathcal{R}_h(B, A) - [J_h(A) < B]$$

$$\Delta_{2h+1}(p) = 2A(H - A) + b(H - 1 - 2A) + B - 2\mathcal{R}_h(B, A) - [J_h(A) < B]$$

Here, the coarse term $A(H - A)$ is the quadratic fold count of complete high-half rows, and the fine placement is supplied by the bit-reversal rectangle counter $\mathcal{R}_h(u, v) = \#\{b' < u : J_h(b') < v\}$, which evaluates in h arithmetic steps as a finite digit sum over the bits u_j of u (read MSB to LSB, with $\rho_0 = 0$ and $\rho_{j+1} = \rho_j + u_j 2^j$):

$$\mathcal{R}_h(u, v) = \sum_{j=0}^{h-1} u_j \left\lceil \frac{v - \rho_j}{2^{j+1}} \right\rceil, \qquad \mathcal{R}_h(H, v) = v$$

Applying this formula to the first 64 bits of π (0x243F6A8885A308D3) resolves its exact position in the 2^{64}-word ring in a single 32-bit pass: $\mathrm{pos}_{64}(\pi) = 2{,}427{,}008{,}858{,}123{,}944{,}441$, with its partner at $16{,}019{,}735{,}215{,}585{,}607{,}174$.

Component 3.4: Spherical Abelianization, Walsh Parity Transport, and Pullback Duality

The Discrete Sphere and Value-Half Latitude Walk

Container CNT-03 embeds the planar rings across all widths into a single three-dimensional spherical coordinate system: radius $r = n$ (word width), polar latitude $z(x) = \mathrm{wt}(x) - \frac{n}{2}$ (with $Z = 2z$ and transverse radius $\rho = \sqrt{n^2 - 4z^2}$), and seam-centered azimuth $\theta_i = 2\pi\left(\frac{i+1/2}{2^n} - \frac{1}{2}\right)$. Projecting the free monoid $S[k, n]$ of k^n ordered interior routes onto its order-forgetting commutative surface (abelianization) collapses binary words ($k = 2$) onto $n + 1$ latitude bands of size $\binom{n}{w}$ (and $\binom{n+k-1}{k-1}$ surface points for general k). An interior reader can reconstruct this global surface area without touching the exterior via the exact Horvitz–Thompson reciprocal fiber identity:

$$\sum_{x \in S[k,n]} \frac{1}{|\mathrm{fiber}(x)|} = |X/\sim| = \binom{n + k - 1}{k - 1}$$

On the sphere, the partner involution P acts as an exact Cartesian reflection (**Theorem 5**, SEALED under 4277c3f1…):

$$P_{\mathrm{nonpal}}(X, Y, Z) = (X, -Y, Z) \quad (\theta \mapsto -\theta,\ z \mapsto z), \qquad P_{\mathrm{pal}}(X, Y, Z) = (X, -Y, -Z) \quad (\theta \mapsto -\theta,\ z \mapsto -z)$$

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Thus, P is a reflection across the XZ-plane on non-palindromes and a $180°$ half-turn about the X-axis on palindromes (equaling the true geometric antipode $(-X,-Y,-Z)$ only at the poles 0^n and 1^n where $\rho = 0$). An azimuthal half-turn $i \mapsto i + N/2 \pmod N$ is provably impossible at every width $n \geq 2$ because $N - 1 - i \equiv i + N/2 \pmod N \Leftrightarrow 2i \equiv N/2 - 1 \pmod N$, which has no integer solution since $N/2 - 1$ is odd for all $n \geq 2$.

In mirror-lifted coordinates (v_j, δ_j), **Proposition 5.1** (SEALED under 359b4d3e…) proves that latitude reads exclusively the **value half** (the agreeing pairs where $\delta_j = 0$), because every disagreeing pair ($\delta_j = 1$) contains one 0 and one 1, contributing zero net weight relative to the equator:

$$z(x) = \sum_{j:\delta_j=0} (2v_j - 1) + [n \text{ odd}]\left(x_h - \frac{1}{2}\right) \quad \Rightarrow \quad |z(x)| \leq h - |\delta(x)| + \frac{[n \text{ odd}]}{2}$$

Consequently, the poles $z = \pm n/2$ belong exclusively to the palindromes 0^n and 1^n, and the more a word differs from its reverse ($k = |\delta|$), the closer to the equator it is confined. Along the $L \to L + 1 \to L + 2$ route, latitude executes a half-step walk: $z_{\text{odd}} = z_{\text{even}} + (m - \frac{1}{2})$ and $z_{\text{child}} = \tau\left[z_{\text{odd}} + (m - \frac{1}{2})(-1)^d\right]$, where $\tau = -1$ iff the core is a canonical palindrome and $m = 1$.

Walsh Parity Split, Spectral Transport, and Exact Spectra of the Comparator and Imbalance

Because the reflected partner address is the bitwise complement $N - 1 - i = i \oplus (N - 1)$, and complementing the input bitvector multiplies the Walsh character $\chi_s(i) = (-1)^{s \cdot i}$ by $(-1)^{|s|}$, the ring ordering diagonalizes the partner involution P into Walsh degree parity (**Theorem 6**, SEALED under be9c1901…). Any P-even word function ($g \circ P = g$) transported to ring addresses $g_R(i) = g(R_n[i])$ has exclusively even-degree Walsh coefficients ($\mathcal{W}[g_R](s) = 0$ for $|s|$ odd), whereas any P-odd word function ($g \circ P = -g$) has exclusively odd-degree Walsh coefficients ($\mathcal{W}[g_R](s) = 0$ for $|s|$ even).

Applying Theorem 6 to the sphere latitude $z(i)$ decomposes the waveform into an orthogonal sum $z(i) = E_n(i) + O_n(i)$: the **Walsh-even body** $E_n(i) = z_+(i)$ supported strictly on non-palindromes, and the **Walsh-odd carrier** $O_n(i) = z_-(i)$ supported strictly on palindromes. By **Corollary 6.1**, the exact fraction of total Walsh energy residing in the odd-degree palindrome carrier is:

$$\frac{E_{\text{odd}}}{E_{\text{total}}} = \begin{cases} 2^{1-n/2}, & n \text{ even}, \\ \dfrac{4h+1}{(2h+1)2^h}, & n = 2h+1 \text{ odd}. \end{cases}$$

In half-word coordinates $u \in \{0, \ldots, 2^{\lceil n/2 \rceil} - 1\}$, the palindromes occur in binary counting order $0,1, \ldots, 2^m - 1$ (making literal R_m order copy a DEAD END for $n \geq 3$), and the compressed carrier has purely linear Walsh degree 1: $2z_{2m}(B_{2m}(u)) = -2\sum_{j=1}^{m} \chi_j(u)$. All higher odd Walsh degrees $(1,3,5, \ldots, n-1)$ in the ring-address carrier arise from the nonlinear placement map S_n via the isometric spectral transport operator $T_n = H_N S_n H_M = \frac{C}{\sqrt{NM}}$ ($T_n^{\top} T_n = I_M$, $C_{s,t} = 0$ when $|s| \not\equiv |t| \pmod 2$), which yields an exact two-stage factored filter $b = \frac{A^{\top} F}{NM}$, $F_{\text{carrier}} = Ab$ using only $m = \lceil n/2 \rceil$ singleton columns.

Over word-value bits x, both the comparator $c_n(x)$ and the prefix imbalance $\Delta_n(p)$ have exact closed-form Walsh spectra (**Theorems 8.1 & 8.2**, SEALED under 21895d2d… and 2f7d21a8…):

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$$c_n(x) = \sum_{j=0}^{h-1} 2^{-j-1} \left(\chi(x_j) - \chi(x_{n-1-j})\right) \prod_{k<j} (1 + \chi(x_k)\chi(x_{n-1-k}))$$

The comparator spectrum has $\#\mathrm{supp}(\hat{c}_n) = 2(2^h - 1)$ nonzero modes (all of odd degree, with $2\binom{h}{k+1}$ modes of degree $2k+1$ and unnormalized coefficients $\pm N/2^{j+1}$). Prefix summation carries $\mathcal{W}[c]$ into $\mathcal{W}[\Delta]$ via a sparse kernel indexed by $j = \mathrm{LSB}(s)$ and $e_k = 2^k$:

$$\mathcal{W}[\Delta](s) = \begin{cases} -\frac{1}{2}\mathcal{W}[c](s), & s \neq 0,\ |s| \text{ odd}, \\ -2^{j-1}\mathcal{W}[c](s \oplus e_j) + \sum_{k=0}^{j-1} 2^{k-1}\,\mathcal{W}[c](s \oplus e_k), & s \neq 0,\ |s| \text{ even}, \\ \frac{N(4N - 3n - 4 - (n \bmod 2))}{24}, & s = 0. \end{cases}$$

Seam Digital Staircase, Closure Digit Sum, and Pullback Duality

At the two boundaries of the Walsh-even body $z_+(i)$ (**Theorems 9.1, 9.2, 9.3,** SEALED under 947b3f56… and 5ba90e6f…), the profile is governed by exact binary digit sums (wt) rather than smooth continuum curves. At the closure ($\theta = \pm\pi$, **Theorem 9.1**), every word $0 \le i < H$ satisfies $R_n[i] = i$, so $z(i) = \mathrm{wt}(i) - n/2$ and $z(N-1-i) = z(i)$ for $1 \le i < H$. Removing the palindrome pole jump leaves $z_+(0) = 0$ and $z_+(1) = 1 - n/2$, creating a normalized step $|e_n(1) - e_n(0)| = 1 - \frac{2}{n}$ across an angular slot of $2\pi/2^n$ and falsifying uniform Lipschitz smooth closure (DEAD END). At the seam ($\theta = 0$, **Theorem 9.2**), in block $K = H - 1 - A$, ordering intra-block offsets $r_i \in \{0, \dots, K-1\}$ by h-bit reversal yields the exact north-pole deficit $\frac{n}{2} - z_+(\mathrm{seam} \pm q) = \mathrm{wt}(K) + \mathrm{wt}(r_i) + (1-b)[n \text{ odd}]$. This forms a non-monotone digital staircase bounded by $2\mathrm{bitlength}(K) - 1 + [n \text{ odd}]$ with mean $\sim \frac{1}{2}\log_2 q$, which returns to deficit 1 at every $K = 2^t$ ($q = K(K-1)/2$). Across the entire interior body (**Theorem 9.3**), $z_+(i) = [a_i \neq J(a_i)] \cdot (\mathrm{wt}(a_i) - n/2)$, proving that the whole body is a single centered digit sum read through the inverse of the Δ rank map.

Pulling the entire structure back (**Proposition 10.1 & Theorem 10.2,** SEALED under 359b4d3e…) exposes a profound coordinate duality between the two readers of the mirror-lifted word. Over word bits x, the value reader (latitude z) and the difference reader (comparison c) are strictly orthogonal ($\sum_x f\,(\mathrm{wt}(x))c(x) = 0$), with z requiring only n singleton Walsh modes and c requiring $2(2^h - 1)$ modes. Transported to ring addresses i, the roles invert completely:

$$c_R(i) = c(R_n[i]) = \chi_{\mathrm{top}}(i)(1 - \mathrm{pal}(i) - 2[i \in \{M-1, M\}]), \qquad \frac{E(\chi_{\mathrm{top}})}{E_{\mathrm{total}}(c_R)} = \frac{(N - 2^{\lceil n/2 \rceil} - 4)^2}{N(N - 2^{\lceil n/2 \rceil})} \to 1$$

The ring is the exact coordinate system constructed so that the comparison of a word with its reverse collapses onto **one single address bit** (χ_{top}, carrying 98.2% of the energy at $n = 12$ and 99.9% at $n = 20$), while the latitude spreads across $1{,}036{,}432$ of the $1{,}048{,}576$ Walsh modes at $n = 20$.

Component 3.5: Circular Tape Transducers, Rolling DI Stacks, and Galois Core Radix Machines

The 2-Bit Ring Rolling Stack and Mod-255 Frobenius Orbit

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Container CNT-04 (two_bit_circle.xlsx, DI_chain_rollup_and_half_ring.ipynb) decompiles the 8-slice circular tape of Ring 2. At width $n = 2$, the four words 00, 10, 01, 11 concatenate clockwise into the tape $T_2 =$ 00100111; read in reverse orientation with 0 and 1 complemented (the "sheet" reading $S_L = C(J(T_L))$), the 8-slice source cycle is $s = (0,0,0,1,1,0,1,1) =$ 0x1B $= 27_{10}$. Sliding a reader window of width $L \in \{2, \dots, 8\}$ starting at slice $i \in \{0, \dots, 7\}$ defines $W_{L,i} = \sum_{k=0}^{L-1} s_{(i+k) \bmod 8}\, 2^{L-1-k}$, which obeys two coupled two-provider recurrences:

$$\text{Radial Growth (Width } L \to L+1\text{):} \quad W_{L+1,i} = 2W_{L,i} + s_{(i+L) \bmod 8} = 2^L s_i + W_{L,i+1}$$

$$\text{Angular Sliding (Phase } i \to i+1\text{):} \quad W_{L,i+1} = 2W_{L,i} - 2^L s_i + s_{(i+L) \bmod 8}$$

Comparing the two updates yields the cross-width fold law $W_{L+1,i} = W_{L,i+1} \Leftrightarrow s_i = 0$. Because $s_0 = s_1 = s_2 = 0$, angular rotation at width L folds directly into radial compilation at width $L+1$, generating the initial spine 0,0,1,3,6,13,27. Moreover, because $\mathrm{rev}(s) =$ 11011000 $= \mathrm{rot}_3(s)$, the counter-clockwise column G is identically the clockwise column B shifted by three slices ($G_{L,i} = B_{L,(i+3) \bmod 8}$ and $G_0^W = B_0^{W+3}$). At full circumference $L = 8$, the entering bit equals the exiting bit ($s_{(i+8) \bmod 8} = s_i$), reducing the step gap $\Delta_i(8) = W_{8,i+1} - W_{8,i}$ to the piecewise linear law $+W_{8,i}$ (if $s_i = 0$) or $W_{8,i} - 255$ (if $s_i = 1$), which is exact doubling modulo $2^8 - 1 = 255$:

$$W_{8,i+1} = 2W_{8,i} - 255 s_i \equiv 2W_{8,i} \pmod{255} \quad \Rightarrow \quad 27 \to 54 \to 108 \to 216 \to 177 \to 99 \to 198 \to 141 \to 27$$

In hexadecimal (1B, 36, 6C, D8, B1, 63, C6, 8D), clockwise and counter-clockwise readings form four bit-reversal pairs (1B ↔ D8, 36 ↔ 6C, B1 ↔ 8D, 63 ↔ C6), and adjacent 8-bit windows satisfy the half-turn nibble swap $W_{8,i} = 16W_{4,i} + W_{4,i+4}$ (1B36 ↔ B163, 6CD8 ↔ C68D). When packing two adjacent L-bit windows via $Q_{L,i} = 2^L W_{L,i} + W_{L,i+1}$, width $L = 5$ marks the first paired-field rollover beyond nibble alignment: for $W_{5,2} =$ $01101_2 =$ 0xD and $W_{5,3} = 11011_2 =$ 0x1B, the 10-bit packed integer is $0110111011_2 = 443_{10} =$ 0x1BB, whereas string concatenation of hex labels produces D1B (3355_{10}).

The Boolean Möbius Transducer Bridge

To link the circular tape directly to the Boolean Möbius transform (ANF difference coordinates) over any power-of-two tape length $N = 2^m$, define $a = Mx$ over $\mathbb{F}_2^N$ by the Pascal-mod-2 subset sum $a_i = \bigoplus_{t \subseteq i} x_t = \bigoplus_{j=0}^{i} \left(\binom{i}{j} \bmod 2 \right) x_j$, where $M^2 = I$. By Lucas's theorem and Pascal's identity $\binom{i}{j} \oplus \binom{i+1}{j} \equiv \binom{i}{j-1} \pmod 2$, conjugating the single-step circular shift $(Rx)_i = x_{(i+1) \bmod N}$ by M yields the exact local nearest-neighbor XOR difference operator $T = MRM = I + S$ (where S is the nilpotent superdiagonal shift, $S^N = 0$):

$$a'_i = (Ta)_i = \begin{cases} a_i \oplus a_{i+1}, & 0 \le i < N-1, \\ a_{N-1}, & i = N-1, \end{cases} \qquad T^N = (I+S)^{2^m} = I + S^{2^m} = I, \qquad a_i = \bigoplus_{j=i}^{N-1} a'_j$$

Here $a_{N-1} = \bigoplus_{j=0}^{N-1} x_j$ is the invariant global tape parity. Simultaneously, tape reversal $(Jx)_i = x_{N-1-i}$ conjugates to the upper superset-XOR operator $U = MJM$ defined by $(Ua)_i = \bigoplus_{j \supseteq i} a_j$ ($U^2 = I,\ UTU = T^{-1}$), bitwise complement $x \mapsto x \oplus 1^N$ toggles only the constant coefficient a_0 ($M(x \oplus 1^N) = Mx \oplus e_0$),

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and the conditional partner P commutes via $MP = \tilde{P}M$, where $\tilde{P}(a) = Ua$ if $Ua \neq a$ else $a \oplus e_0$ (verified across $256/256$ bytes and $32{,}768/32{,}768$ window reads).

Reader Steps, Fermat Register Primes, the Join Theorem, and BBP Radix Dynamics

Decompiling base representations across the 64-bit register stack (One Program, Infinite Compilers) proves that a numeral base adds zero distinctions to the substrate; a base is strictly the reader's block step width (steps × width = LEN). In 4,000 seeded 64-bit single-bit-flip trials, power-of-two block bases ($b = 2,4,8,16,256$ with $w = 1,2,3,4,8$ bits) alter an average of 1.00 digit (max 1) per bit flip, whereas non-power-of-two bases are global conversions that cascade via carries across 14.97 digits in Base 3 (max 35), 9.48 digits in Base 10 (max 20), and 7.66 digits in Base 22 (max 15). On the invertible Baker's map $B_b(x, y) = (\{bx\}, (y + \lfloor bx \rfloor)/b)$, four Base-2 steps and one Base-16 step produce the identical expanding coordinate x ($2{,}000/2{,}000$ rational trials), while the collapsing history coordinate y records the 4-bit nibble in reversed bit order ($a_0/16 + a_1/8 + a_2/4 + a_3/2$ vs. $a_0/2 + a_1/4 + a_2/8 + a_3/16$), agreeing only on the four palindromic nibbles 0000, 0110, 1001, 1111.

Decompiling the register stack from 64 bits downward via the Burnside–Möbius necklace identity $2^n = \sum_{d|n} d \cdot P_2(d)$ with $P_2(d) = \frac{1}{d}\sum_{k|d} \mu(k) 2^{d/k}$ shows that smaller registers ($d \in \{1,2,4,8,16,32\}$) sit inside the 64-bit ring as the exact sets of periodic words fixed by half-turn rotations ($uu \in \mathbb{F}_{2^{n/2}}$). Doubling the register width multiplies the multiplicative group order by a Fermat factor $2^{2m} - 1 = (2^m - 1)(2^m + 1)$, introducing only Fermat-divisor primes ($2^{64} - 1 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 6700417$, where $F_5 = 2^{32} + 1 = 641 \times 6{,}700{,}417$) while excluding odd-order primes such as $7,11,13,19,23,29,31$.

In §133 (**The Join Theorem**), an n-lane radix machine in base $b = p^{n/2}$ evaluates linear combinations of $L_j = \sum_{k=0}^{\infty} b^{-k}/(nk + j)$, whose wheel returns $\beta_m = \sqrt{p} - \zeta_n^m = \sqrt{p}(1 - w_m)$ ($\zeta_n = e^{2\pi/n}$, $w_m = \zeta_n^m/\sqrt{p}$) factor the register ring:

$$\prod_{m=0}^{n-1} \beta_m = \prod_{m=0}^{n-1} (\sqrt{p} - \zeta_n^m) = p^{n/2} - 1$$

A return β_m is prime-free (an algebraic unit) in only two cases: **scale returns** ($\zeta_n^m = \pm 1$), where $|N(\sqrt{p} \mp 1)| = p - 1 = 1 \Leftrightarrow p = 2$, and **phase returns** ($|\sqrt{p} - \zeta_n^m| = 1 \Leftrightarrow \mathrm{Re}(w_m) = 1/2$, the 0.5 potential midpoint), which occur only for $p = 2$ ($n = 8$, $w_1 = \frac{1+i}{2} = s$, $\sqrt{2} - \zeta_8 = \zeta_8^{-1}$) and $p = 3$ ($n = 12$, $w_1 = \frac{1}{2} + \frac{\sqrt{3}}{6}i$, $\sqrt{3} - \zeta_{12} = \zeta_{12}^{-1}$). Thus, binary ($p = 2$) is the unique prime where scale and phase returns meet as prime-free units ($a_0 + a_{n/2} = 2a_{n/8}$). By Baker's theorem and Galois descent, the rational zero-route space is the Galois-stable core $Z_p(n) = \dim R_0 = \dim \bigcap_{\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_n, \sqrt{p})/\mathbb{Q})} \pi_\sigma(R)$ (explaining why off-chain $n = 12$ drops from $d_R = 2$ to $Z_2(12) = 1$ under σ_7, and $n = 24$ drops from 7 to 5, while on the binary register chain $n = 2^e$, $Z_2(2^e) = 1$ is spanned by the pullback of the 8-lane BBP zero route).

In the 8-lane Bailey–Borwein–Plouffe (BBP) machine $\pi = 4L_1 - 2L_4 - L_5 - L_6$ (Nest.docx), every lane phase $x_{k,j}(n) = \{16^{n-k}/(8k + j)\}$ is compiled at step 0 ($S_0 = \pi - 4$) and evolves simultaneously via $x_{k,j}(n + 1) = \{16x_{k,j}(n)\}$, with its first crossing $q_{k,j}(n) = \lfloor 16x_{k,j}(n) \rfloor$ occurring at $n = k + \lceil \log_{16}(8k + j) \rceil - 1$. Because the weights $c = (4, -2, -1, -1)$ satisfy $\sum c_j = 0$ and $\sum j c_j = -15$, the rational layer value

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compresses to $V_k = \frac{120k^2+151k+47}{(8k+1)(2k+1)(8k+5)(4k+3)} = \frac{15}{64k^2} + O(k^{-3})$, delaying the first nonzero hexadecimal digit $w_k(m)$ by a 2-to-3 step **silent carry band** governed by the bounded internal carry $J_k(m) \in \{-4, \dots, 3\}$:

$$H_k(m) = w_k(m) + 16J_k(m) - J_k(m+1), \qquad \sum_{m=0}^{\infty} H_k(m) 16^{-m-1} = V_k$$

Behind the real line, the quarter-turn step $s = \frac{1+i}{2} = 2^{-1/2} e^{i\pi/4}$ contracts area by $\det(M_s) = |s|^2 = 1/2$ (1 bit/tick, $s^4 = -1/4,\ s^8 = 1/16$) and splits $-\log(1-s) = \frac{1}{2}\log 2 + i\frac{\pi}{4}$ into symmetric real ($\log 2 = \int_0^1 K(x)\,dx$) and antisymmetric imaginary ($\pi = \int_0^1 \frac{4K(x)}{1-x^2}\,dx$) shadows of the kernel $K(x) = \frac{2(1/2-x)}{|1-e^{i\pi/4}x|^2}$, which changes sign precisely at the $x = 1/2 = |s|^2$ crest.

Component 3.6: Homological Cut Algebra, Token Arenas, and Carry-Capacity Semigroups

The Token-Arena Cohomological Tower and Universal Descent Identity

Container CNT-05 (The Ceiling and the Descent, Two Quotients of One Machine) formalizes "select first, transform second" across multi-token graph arenas $\mathcal{A}(X, K)$:

$$X \to \mathcal{A}(X,K) \to C^1/\mathrm{im}d_0 \to R_G(X,K) \to \mathrm{Im}[T_p]_G \to \text{Accessible Observation}$$

Across successive relational orders—Order 0 (state orbits C^0), Order 1 (transitions modulo exact potentials $C_G^1/\mathrm{im}d_0$), Order 2 (route cycles $\ker d_1$ modulo declared 2-cells K_{disj} or K_4), and Order 3 (relations among relations via commuting 3-cubes $\partial[A,B,C] = [B,C]_{A=1} - [B,C]_{A=0} - [A,C]_{B=1} + [A,C]_{B=0} + [A,B]_{C=1} - [A,B]_{C=0}$ with $d_2 d_1 = 0$)—Hodge duality yields the **Universal Descent Identity** (verified across 20/20 Order-1 and 12/12 Order-2 arenas):

$$\varepsilon_{\text{harm}} = \dim Z - \dim\text{span}(\text{declared cells})$$

This identity separates zero readings into three distinct structural regimes: genuine cycle absence ($\varepsilon = 0$, e.g., paw), route-order closure ($\varepsilon > 0, \varepsilon_{\text{harm}} = 0$, e.g., wheel6 where 3 channels close under K_{disj}), and single-order screening ($\varepsilon_{\text{harm}} = 0, \varepsilon_{\text{harm}}^2 = 1$, e.g., prism at $K = 3$, where an Order-1 reader misses the surviving Order-2 channel).

Two Orthogonal Quotients of Binary Addition and Backward Non-Invertibility

When modeling the n-bit binary adder as a transition system, the specification separates two non-interchangeable quotients that collapse 2^{2n} states to polynomial carry-seam invariants. The first invariant is the **Behavioral Myhill–Nerode Cut Complexity** $F(S)$: for a subset of read bit positions $S \subseteq \{0, \dots, n-1\}$, the minimal forward-closed reader state is the coupled pair (S, Φ_S), which updates locally via $U(\Phi_S(\alpha), S, p, u)$ across all 11 geometric update classes (75,000 updates verified at $n = 6$). At the boundaries, $F(\varnothing) = 1$ and $F([0, n-1]) = 1$, while intermediate cuts satisfy $F(S) \geq 2$ (1,2,2, …,2,1 along the optimal minimax LSB-first schedule). By **Theorem A"**, forward restriction is automatically functional ($\Phi = \mathrm{MN} \Longrightarrow$ forward congruence), whereas backward "un-reading" is non-functional for all $n \geq 3$. Specifically, at the bottom-anchored cut $S = \{0,1\}$, two histories α ((0,0) kill at bit 0, (0,1) transparent at bit 1) and β ((0,0) kill at bit 0, (0,0) kill at bit 1) merge into $\Phi_{\{0,1\}}(\alpha) = \Phi_{\{0,1\}}(\beta) = (\text{bot}, 0)$, but un-reading bit 0

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to reopen $S' = \{1\}$ exposes bit 1 to a potential carry and splits them into $\Phi_{\{1\}}(\alpha) = (\text{trans}) \neq \Phi_{\{1\}}(\beta) = (\text{int}, 1, 0)$.

The second invariant is the **Homological Disjoint-Square Residual** $b_1^{\text{disj}}(n)$: modeling the adder as a cubical complex generated by increments $+2^i$ and $+2^j$ $(i < j)$, the commutativity square at basepoint x is declared local if and only if $D_{ij}(x) = 1 \Leftrightarrow$ bits $i, \ldots, j-1$ of x are not all 1. Downward induction via low-bit transport (Lemma L1) and staircase decomposition (Lemma L2) proves that the undeclared pair module M_{ij} has exact dimension $\dim M_{ij} = n - j$ (the number of carry-run continuation depths above seam j), yielding the global cohomology rank:

$$b_1^{\text{disj}}(n) = \binom{n}{2} + (n-1) = \frac{(n-1)(n+2)}{2}, \qquad \dim(B_A/B_D) = \binom{n}{2}$$

The Conditional Cut Law and Constant-Transfer Semigroup Threshold

In Carry_Capacity_and_the_Conditional_Cut_Law, the condition under which an unread gap G of width g severs dependence between neighboring read runs is proven to be a **transfer-semigroup controllability law**, not a storage-cardinality law. Let $\mathcal{C}$ be the crossing carry alphabet and $\mathcal{T}_G \subseteq \text{End}(\mathcal{C})$ be the family of realizable transfers across G. Having $|\mathcal{T}_G| \geq |\mathcal{C}|$ does **not** imply separation (e.g., $(b, m, g) = (2,3,1)$ realizes 4 distinct maps on $|\mathcal{C}| = 3$ states with zero constants; $(2,8,2)$ realizes 25 maps on 8 states with zero constants). Instead, separation holds if and only if $\mathcal{T}_G$ contains a **constant transfer map**:

$$\exists \tau \in \mathcal{T}_G \quad \text{such that} \quad \tau(c) = c_* \quad \forall c \in \mathcal{C}$$

For m-operand addition in base b ($\mathcal{C}_m = \{0, \ldots, m-1\}$), the all-zero gap word implements repeated integer division $\tau_{0^g}(c) = \lfloor c/b^g \rfloor$, which is optimal across all 84 tested triples and collapses $\mathcal{C}_m$ to 0 if and only if:

$$\left\lfloor \frac{m-1}{b^g} \right\rfloor = 0 \quad \Leftrightarrow \quad b^g \geq m \quad \Leftrightarrow \quad g \geq \lceil \log_b m \rceil$$

Component 3.7: Symbolic Cofactor Routing and Multiscale Transduction in SHA-256

Phase Lenses, Round Bijections, and the Failure of Local DAG Propagation

Container CNT-06 (Algebraic Routing.docx, ChatGPT_SHA-Decoder-Analysis) applies the DI subtraction discipline to SHA-256. First, on individual 32-bit words, the 18-spoke phase lens $q(x) = 18x \bmod 2^{32}$ aliases words differing in their MSB ($q(x) = q(x \oplus 2^{31})$), but retaining the MSB as an orthogonal qualifier restores an exact two-coordinate decoder across $100{,}192/100{,}192$ tests:

$$x_{\text{low}} = \left(\frac{q(x)}{2} \cdot 954437177\right) \bmod 2^{31}, \qquad x = x_{\text{low}} \;\mid\; (\text{MSB}(x) \ll 31)$$

Second, within the 64-round compression chain, subtracting the initial vector H_0 from the digest D via Davies–Meyer feed-forward inversion ($Z = D - H_0 \pmod{2^{32}}$) factors cleanly into 8 independent 32-bit words. When the message schedule W_t and round constants K_t are supplied as drivers, the 8-register round update $(a, b, c, d, e, f, g, h) \mapsto (T_1 + T_2, a, b, c, d + T_1, e, f, g)$ is an exact bijection (verified across $10{,}000$ random trials), and differential carrier-locked traces satisfy exact modular closure across all 64 rounds ($\Delta T_1 = \Delta T_{1,\text{partial}} + \Delta W$, $\Delta a_{\text{next}} = \Delta T_1 + \Delta T_2$, $\Delta e_{\text{next}} = \Delta d_{\text{before}} + \Delta T_1 \pmod{2^{32}}$).

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To invert the unknown message bits μ without candidate enumeration, shared_carry_inverse.py (sealed under 6b9a7d65…) compiles all 64 reverse rounds into a shared symbolic Directed Acyclic Graph (DAG) over $\{\oplus,\wedge,1\}$ using the exact bit-level carry and borrow factorizations ($p_j = x_j \oplus y_j,\ g_j = x_j y_j,\ s_j = p_j \oplus c_j$):

$$c_{j+1} = g_j \oplus p_j c_j = \oplus_{i=0}^{j} g_i \prod_{k=i+1}^{j} p_k \ \oplus\ c_0 \prod_{k=0}^{j} p_k$$

When lifted from $\mathbb{F}_2$ to $\mathbb{Z}$ via PSLQ integer-relation recovery and 8-row Möbius inversion over $(1, x, y, xy, c, xc, yc, xyc)$, the single-bit carry and borrow gates become exact pointwise integer polynomials:

$$\mathrm{carry}(x,y,c) = xy + xc + yc - 2xyc, \qquad \mathrm{borrow}(x,y,c) = y + c - xy - xc - yc + 2xyc$$

Doubling the unknown message interface from 1 byte (Q, 0x51, 8 bits) to 2 bytes (GO, 16 bits) increases unique DAG nodes by only 0.90% ($117{,}240 \to 118{,}294$ nodes in ~ 0.14 s). However, tree-local unit propagation binds $2{,}717$ and $2{,}247$ internal nodes while binding **zero message bits** ($0/8$ and $0/16$), because correlation constraints across Ch, Maj, σ_0, σ_1, and carry chains are distributed across reconvergent nonlinear fanout loops.

ROBDD Shannon Cofactor Collapse, Schedule-Tail Bijection, and the 184-Bit Selector Bound

Resolving reconvergent nonlinear loops requires compiling the symbolic DAG into a Reduced Ordered Binary Decision Diagram (ROBDD) via recursive Shannon cofactor conjunction (cofactor_inverse_bdd.py):

$$F = (1-\mu_j)F_0 + \mu_j F_1, \qquad G = (1-\mu_j)G_0 + \mu_j G_1 \quad \Rightarrow \quad F \wedge G = (1-\mu_j)(F_0 \wedge G_0) + \mu_j (F_1 \wedge G_1)$$

Because recursive cofactoring forces every reconvergent path sharing μ_j to assume the identical truth value simultaneously inside $F_0 \wedge G_0$ and $F_1 \wedge G_1$, intersecting the 1-byte inverse graph with the required initial state H_0 (Word 0 = 0x6A09E667, LSB first) annihilates all incompatible assignments in just **7 bit constraints** ($256 \to 131 \to 56 \to 28 \to 12 \to 6 \to 3 \to 1$), isolating the unique characteristic polynomial:

$$\Phi_H(\mu) = \mu_0(1 \oplus \mu_1)(1 \oplus \mu_2)(1 \oplus \mu_3)\mu_4(1 \oplus \mu_5)\mu_6(1 \oplus \mu_7) \quad \Rightarrow \quad \mu = 01010001_2 = 0x51\ (\text{'Q'}) \\ = R_8[70]$$

Peak intermediate ROBDD complexity remains invariant within 0.29% ($80{,}556$–$80{,}788$ nodes) across four variable orderings because 64-round SHA-256 diffusion reaches full algebraic degree.

At the macro-coordinate scale, two exact theorems bound the full-block inversion problem. First, under the **2048-Bit Rolling Trace Prefix Law**, concatenating the 64 first-lane updates $U_t = T1_t + T2_t \pmod{2^{32}}$ into $X \in \{0,1\}^{2048}$ maps the digest $Z = D - H_0$ bijectively onto the final 8 words $s(H) = (U_{56}, \ldots, U_{63})$, so applying the ring partner P_{2048} fixes the 256-bit prefix $\mathrm{prefix}_{256}(P_{2048}(X)) = \mathrm{rev}_{256}(s(H))$ (though unconstrained reflection breaks the 48 internal schedule consistency equations). Second, under the **512-Bit Schedule-Tail Bijection and Pigeonhole Selector Bound**, solving the forward schedule recurrence $W_t = \sigma_1(W_{t-2}) + W_{t-7} + \sigma_0(W_{t-15}) + W_{t-16} \pmod{2^{32}}$ for W_{t-16} yields the deterministic backward recurrence:

$$W_{t-16} = W_t - \sigma_1(W_{t-2}) - W_{t-7} - \sigma_0(W_{t-15}) \pmod{2^{32}}$$

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Iterating this backward recurrence from $t = 63$ down to $t = 16$ establishes a lossless three-way bijection between the 16-word schedule tail $T = (W_{48}, \dots, W_{63}) \in \{0,1\}^{512}$, the initial message block $B = E^{-1}(T) = (W_0, \dots, W_{15})$, and the 512-bit ring address $i = \text{location}_{512}(E^{-1}(T))$. However, for the class X of single-block 55-byte payloads ($|X| = 256^{55} = 2^{440}$), the pigeonhole principle forces $\max_D |F^{-1}(D)| \geq 2^{440}/2^{256} = 2^{184}$, proving that any deterministic locator recovering the historical source from D alone requires a retained **source-fiber selector coordinate $r(D)$ of at least 184 bits** (whereas short interfaces $|X| \leq 2^{256}$ generically have singleton fibers $|F_D| = 1$).

Component 3.8: Physical, Atomic, Nuclear, and Biological Substrate Projections

Exact Decompilation of 3D Angular Momentum and the Periodic Table

In One Program, Infinite Compilers and The Reading of Potential, the capacities of atomic electron shells and nuclear magic numbers are decompiled directly from the doubled binary ring without external quantum-number postulates. Because a relational difference requires two frames ($C0'$), a subshell component of radial size ℓ is read across a two-position gap as a doubled binary word of length 2ℓ bits. Projecting this 2ℓ-bit ring onto its order-free commutative surface yields $2\ell + 1$ discrete weight latitudes $w \in \{0, \dots, 2\ell\}$. Constructing the collective Schwinger spin operators S_x, S_y, S_z from the 2ℓ two-state bits proves in exact rational arithmetic (verified for $\ell = 1,2,3,4$) that the symmetric order-free layer carries the exact spin-ℓ representation:

$$S^2 = S_x^2 + S_y^2 + S_z^2 = \ell(\ell+1)I, \qquad \text{Capacity}(n) = \sum_{\ell=0}^{n-1} 2\,(2\ell+1) = 2n^2 \Longrightarrow 2,8,18,32,50,72,98$$

Here the factor of 2 is the 1-bit pass/flip reader step (Wootters–Fields $d = 2$), and the admissibility rule $\ell < n \Leftrightarrow n_r = n - \ell - 1 \geq 0$ expresses the domain law that no proper part equals the whole. Traversing the 2-generator surface by Madelung degree $d = n + \ell$ ($d = 1, \dots, 10$), a nodeless orbital ($\ell = n - 1 \Longrightarrow n_r = 0$) has odd degree $d = 2n - 1$ ($1s, 2p, 3d, 4f$ at $d = 1,3,5,7$). Its hypothetical predecessor one slice lower at even degree $d - 1 = 2n - 2$ would require quantum numbers $(n-1, n-1)$ ($0s, 1p, 2d, 3f$) with $n_r = -1 < 0$, which is inadmissible (no_way_home.py). Consequently, every even slice $d = 2k$ adds zero new angular momentum types and must repeat the exact capacity of $d = 2k - 1$, forcing the closed-form period-doubling law (SEALED under 478d1baf):

$$C(d) = 2\left\lfloor \frac{d+1}{2} \right\rfloor^2 \Longrightarrow 2,2,8,8,18,18,32,32,50,50 \quad \overset{\text{cut at new } s}{\rightarrow} \quad Z_{\text{noble}} = 2,10,18,36,54,86,118$$

In the nuclear shell model (nuclear_unfold.py), relativistic spin-orbit coupling advances rather than delays the partnerless top-j orbital of each harmonic oscillator shell (2,8,20,40,70,112), dropping it into the lower shell to produce the magic numbers 28,50,82,126,184 (reported against 954 ENSDF first-2^+ records and 675 even-even nuclei at $p = 2.11 \times 10^{-6}$ neutron and 5.29×10^{-6} proton significance) and algebraically resolving the subshell closure at $14 + 2 = 16 \rightarrow 20$.

Interpretive Physical and Biological Bridges in the PULL Sector

In strict compliance with the ledger discipline, broader physical and biological projections are documented as PULL bridges—meaning their discrete compiler invariants RUN, while their ontological identification with

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physical observables remains an open hypothesis requiring an independent dimensional map. On the bit-sphere, the North and South poles ($w = 0, n$) have fiber size $\binom{n}{0} = \binom{n}{n} = 1$, zero permutation entropy, and zero internal clock, corresponding to massless excitations propagating along the null cone $ds^2 = c^2dt^2 - dx^2 = 0$ ($d\tau = 0$). Conversely, at the equator of an even ring $2m$ ($w = m$), the fiber reaches its maximum multiplicity $\binom{2m}{m} \sim \frac{2^{2m}}{\sqrt{\pi m}}$, with $\text{Catalan}(m) = \frac{1}{m+1}\binom{2m}{m}$ closed Dyck wrapper objects at the band just above the equator. When spatial drift vanishes ($\nabla\phi = 0$), satisfying C1 continuation forces the equatorial state to oscillate internally across its diameter at the Klein–Gordon Compton/Zitterbewegung frequency $\omega = mc^2/\hbar$ (7.76×10^{20} rad/s for the electron), while in the discrete edge update $J_{ij}(t+1) = \alpha J_{ij}(t) - \beta(\rho_j - \rho_i) - \mu^2(\rho_i + \rho_j)/2$, the mass term $-\mu^2(\rho_i + \rho_j)/2$ breaks shift symmetry by reading state amplitude rather than spatial difference. Similarly, reading Law 3 off the ring identifies the constant second derivative -1 of angle against value ($\frac{1}{2} - u = \frac{1}{2}(1-t)^2$) as the product of mass and acceleration at the seam collision, though assigning which factor is mass and which is acceleration remains OPEN.

In SHA constants as FPGA layout, executing the bitwise triad (XOR = Curvature, AND $\ll 1$ = GAP Residue, SUM = Value Mirror) across fractional prime-root seeds yields normalized GAP ratios between 0.5625 ($18/32$ bits) and 0.6875 ($22/32$ bits), complementary to the 0.4375 thermodynamic tolerance. Across a 64-round compression chain, the persistent non-zero GAP averages 16.0 bits/round (a 50/50 bifurcation between the Observable Value Channel and Recoverable Shape Channel), accumulating $64 \times 16.0 = 1{,}024$ bits of "topological mass" (Informational Gravity) that acts as a self-witnessing Reverse DNS pointer back to the parent input, culminating in the 17.10 Scar entropy floor (analogous to JC-1 mitochondrial membrane potential ratios) and steered by the Mark 1 Attractor ($H \approx \pi/9 \approx 0.349$). In carbon substrates, the framework models DNA base-pairing as a Glass Key compression engine ($40{,}000{:}\,1$ ratio) rendering 3D proteins via an Inverse Fast Fourier Transform (IFFT) at a 33 Hz frame rate, where malignancy corresponds to harmonic decoherence.

Level 4 Code Verification Ledger, Cryptographic Seals, and Frontier Directives

Master Verification Ledger and Routed Dead Ends

The table below consolidates the Level 4 code-level equations, verification scopes, and final ledger classifications across every container of the @reverse-the-gap software architecture.

Domain / Component	Mathematical Formula / Operator Law	Ledger Status	Verified Computational Scope & Provenance
Width-3 Exchange (CNT-01**)**	$F = T \circ X \circ T = (a \oplus ab \oplus c,\ 1 \oplus b,\ a \oplus bc)$, $F^2 = I, \text{Fix}(F) = \varnothing$	SEALED, RUNS	Exact group orders $n = 1{,}2$ ($2! = 2{,}4! = 24$); 8-state ANF degree $n - 1 = 2$ check.

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Domain / Component	Mathematical Formula / Operator Law	Ledger Status	Verified Computational Scope & Provenance
Even-Base Partner (CNT-02)	$P_b^2 = I,\ \mathrm{Fix}(P_b) = \varnothing \Leftrightarrow$ b is even; $N_{\text{maybe}}/b^2 = 1/b$	SEALED, RUNS	Proved algebraically; verified for $b = 2,4,6$ and $b^{\lfloor L/2 \rfloor}$ palindromes.
All-Word Prefix Rank (CNT-02)	$\mathrm{rank}(p) = \frac{1}{2}(p - \pi(p) + \Delta(p)) + \pi_c(p)$	SEALED, RUNS	All $2{,}097{,}148$ thresholds ($n = 2 \ldots 20$); 64-bit π at $2{,}427{,}008{,}858{,}123{,}944{,}441$; widths to $1{,}024$.
Closed Imbalance (CNT-02)	$\Delta_{2h}(p) = A(H - A) + B - 2\mathcal{R}_h(B, A) - [J_h(A) < B]$; $\mathcal{R}_h = \sum u_j \lceil (v - \rho_j)/2^{j+1} \rceil$	SEALED, RUNS	$2{,}097{,}167$ thresholds ($n = 2 \ldots 20$); $88{,}408$ $\mathcal{R}_h$ pairs; 90 wide queries to $n = 1{,}024$.
Borrow Gate (CNT-02)	$c(x) = \mathrm{sgn}(J(x) - x) = \beta(x - J(x)) - \beta(J(x) - x)$, $\beta' = \mathrm{Maj}(\neg a, b, \beta)$	SEALED, RUNS	Exhaustively verified on every word $n = 2 \ldots 16$; $i_5(u) = 4u + 2u_2 + \mathrm{Maj}(1 - u_0, u_1, u_2)$.
Mirror Lift Codec (CNT-02)	$E_m(x) = \mathrm{canon}(mxm)$, $G_m(x) = 0P^m(x)1$; route residuals = $\delta_{h-1} \ldots \delta_0$	SEALED, RUNS	$699{,}048$ E/G outputs ($L = 2 \ldots 18$, target 20); $8{,}188$ cross-bundle words; 512-bit round trips.
Sphere & Latitude (CNT-03)	$P(X, Y, Z) = (X, -Y, \pm Z)$; $z(x) = \sum_{\delta_j = 0} (2v_j - 1) + [n \text{ odd}](x_h - \frac{1}{2})$	SEALED, RUNS	All $2{,}097{,}148$ words ($n = 2 \ldots 20$), 228 weight bands; pole traces to $n = 4{,}096$.
Walsh Parity & Energy (CNT-03)	$g \circ P = \pm g \Longrightarrow$ even/odd Walsh degrees; $E_{\text{odd}}/E_{\text{total}} = 2^{1-n/2}$ (even n)	SEALED, RUNS	Exact rational verification $n = 2 \ldots 20$; isometric transport $C^{\mathsf{T}}C = NMI_M$ ($n = 2 \ldots 12$).

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Domain / Component	Mathematical Formula / Operator Law	Ledger Status	Verified Computational Scope & Provenance
Comparator & Δ Spectra (CNT-03)	$\#\mathrm{supp}(\hat{c}_n) = 2(2^h - 1)$; $2\mathcal{W}[\Delta](s) = -2^j \mathcal{W}[c](s \oplus e_j) + \sum_{k<j} 2^k \mathcal{W}[c](s \oplus e_k)$	SEALED, RUNS	86,868 kernel pairs ($n = 2 \ldots 8$); 8,177 nonzero modes ($n = 2 \ldots 12$); zero mode $n = 2 \ldots 20$.
Seam & Body Laws (CNT-03)	$z_+(i) = [a_i \neq J(a_i)](\mathrm{wt}(a_i) - n/2)$; seam deficit $= \mathrm{wt}(K) + \mathrm{wt}(r_i) + (1 - b)[n \text{ odd}]$	SEALED, RUNS	522,242 paired-face samples; 3,049 blocks (including $K \geq H/2$ palindromes); scalar to 1,025.
Pullback Duality (CNT-03)	$c_R(i) = \chi_{\mathrm{top}}(i)(1 - \mathrm{pal}(i) - 2[i \in \{M - 1, M\}])$; top-bit energy → 1	SEALED, RUNS	Verified $n = 2 \ldots 20$ (98.2% at $n = 12$, 99.9% at $n = 20$); Table 2 participation ratios.
Möbius Tape Bridge (CNT-04)	$MR = TM = (I + S)M,\ T^{2^m} = I,\ MJ = UM,\ MP = \tilde{P}M$	STANDS, RUNS	Proved via Lucas/Pascal mod 2; 256/256 bytes and 32,768/32,768 window reads ($L = 1 \ldots 16$).
Galois Core & BBP (CNT-04)	$Z_p(n) = \dim \bigcap_\sigma \pi_\sigma(R),\ Z_2(2^e) = 1$; $H_k(m) = w_k(m) + 16J_k(m) - J_k(m+1)$	STANDS, RUNS	47/47 Galois core checks (§133); 7,988 BBP lane crossings and 45,030 carry ages.
Adder Cohomology & Cuts (CNT-05)	$b_1^{\mathrm{disj}}(n) = \binom{n}{2} + (n - 1),\ \dim M_{ij} = n - j$; gap reset ⇔ $b^g \geq m$	STANDS, RUNS	All-n proof (Lemmas L1, L2); 75,000 MN updates ($n = 6$); 84 (b, m, g) semigroup triples.
ROBDD & Tail Routing (CNT-06)	$F \wedge G = (1 - \mu_j)(F_0 \wedge G_0) + \mu_j(F_1 \wedge G_1)$; $E: B \leftrightarrow T$; selector ≥ 184 bits	BOUND, RUNS	1-byte preimage 0x51 bound in 7 constraints (80,556 peak nodes); 1,000/1,000 tail round trips.

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Domain / Component	Mathematical Formula / Operator Law	Ledger Status	Verified Computational Scope & Provenance
Atomic/Nuclear Shells (CNT-06**)**	$S^2 = \ell(\ell+1)I$; $C(d) = 2\lfloor (d+1)/2 \rfloor^2 \Longrightarrow$ 2,10,18,36,54,86,118	SEALED, RUNS	Exact rational Schwinger spin ($\ell = 1 \ldots 4$); inadmissibility $n_r = -1$ (478d1baf).
Routed Dead Ends	Literal half-width order; smooth closure; $\frac{1}{2}(p + \pi + \Delta)$ for all reps; LSB-toggle $\widehat{\Delta}$; SHA route codes	DEAD END	Routed respectively to: binary counting order; address digit sum; non-canonical palindromes; full prefix kernel; codec relabeling (parked).

Cryptographic Seal Registry and Execution Provenance

Every computational module in the @reverse-the-gap release pipeline is anchored by its pre-execution SHA-256 digest, providing a tamper-evident audit trail across all executed runs.

Verification Module / Audit Artifact	Reference Script / File	SHA-256 Cryptographic Seal Digest	Provenance Notes & Execution Scope
Width-Three Exchange Compiler	nexus_reverse_gap_width3.py	52eca8c48a887f4fb9235a1a4c297d701a9b83f8af0180a93c8cb25ac6b4506e	Sealed prediction verified prior to group-order and ANF check.
Symbolic	shared_carry_inverse.py	6b9a7d6549cd336c97a884fe85f087e6d67c542254a4c5f443a54df41cccd8b5	Sealed prior to

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Verification Module / Audit Artifact	**Reference Script / File**	**SHA-256 Cryptographic Seal Digest**	**Provenance Notes & Execution Scope**
Carry/Borrow Inverse DAG			linking 1-byte (Q) and 2-byte (GO) target digests.
Two-Branch Odd-to-Even E/G Lift	nexus_odd_even_ring_lift_audit.py	ab543c247d2adcfaad114cf45a023697396fac5bb0fd0dc52a573da0b39a7fb0	Exhaustive partition verification through target width $L + 2 = 20$.
Scalar Branch Position & Seam Map	position_map_audit.py	75ca9d286bc1dfab84a6878ae800f50f7ae54ddf954920a589784e0aec1eb6cf	Verified across $699{,}048$ E/G outputs and widths $64{,}256{,}51$.

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Verification Module / Audit Artifact	**Reference Script / File**	**SHA-256 Cryptographic Seal Digest**	**Provenance Notes & Execution Scope**
Route-to-Rank-to-Address Closure	nexus_route_rank_sphere_audit.py	ee827844ba91d592c81cb1f6f22f42dc0a2ea9782 0a3d532e199d6e30803020b	Cross-bundle verified across 8,188 words at widths $2 \ldots 12$.
Reflection–Lift & Sphere Latitude	reflection_lift_sphere_audit.py	4277c3f1b18df45ab669f00f75e33596458216d315 73983dcec1e2269e607d8f	Verified across 2,097,148 words ($n = 2 \ldots 20$) and pole traces to 4,096.
Walsh Parity Split & Energy Share	walsh_split.py	be9c190152f9d9e0f0cade476691f24aa0153ef9af 083dd41e0b30b8fbfa652c	Exact rational energy share verified for $n = 2 \ldots 20$.

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Verification Module / Audit Artifact	Reference Script / File	SHA-256 Cryptographic Seal Digest	Provenance Notes & Execution Scope
Palindrome Address & Borrow Gate	address_comparator.py	5910642ece837f543f4d363abd5f1951f13c06758be3612dd8e6015712438bfb	Verified across all palindromes ($n = 2 \ldots 20$) and words ($n = 2 \ldots 16$).
Comparator Imbalance & Spectrum	comparator_imbalance_audit.py	5d5583fae11e6e2f75c6ab01ccefd63fb5370367efe9c3063d839294fe86be66	Source seal; prediction without newline 21895d2d… (file c560fad2…).
Even-Delta Kernel & Zero Mode	even_delta_spectrum_audit.py	2f7d21a88107b4147843d550cdd0689f22e197b5760a0fe1050787631fee01c7	Source 982f6838…; second run prediction f3275b7

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Verification Module / Audit Artifact	Reference Script / File	SHA-256 Cryptographic Seal Digest	Provenance Notes & Execution Scope
			b… (source 8c8668 20…).
Seam Block Sums & Pointwise Profile	seam_closure.py	947b3f56802focb601cea3fad19cd8fd7075518fe1 6737f316665105cbe59adb	Pointwise seam prediction 5ba90e 6f… (source 90f5a9e 7…).
All-Word Interior Rank & Body	interior_body_audit.py	5cde4126a76c415bd4397164cf1b09b1a07fcdb82 f40e140fd539a9f065757a1	Source 756fb65 7…; second block-sum run 358e76 d2… (source 215b2a 8b…).
Pullback Duality	pullback_run.py	359b4d3e9e043262928226047a9bed3feb0122a7 b5c672464c11909bcd1584bd	Verified all six pullback

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Verification Module / Audit Artifact	**Reference Script / File**	**SHA-256 Cryptographic Seal Digest**	**Provenance Notes & Execution Scope**
& 64-Bit π Rank			propositions (PB1–PB6) in 9.9 s.
Folded Square & Ten Routes Suite	nested_ring_routes.ipynb	18cbef3f425c5a51c727a59a4dacb9cdc43b6c9bf24771ec5eeace1c0ae7163f	Paired with 721fafeb…, 3cc4adb2…, 4b23106d…, and 325c8204….

Open Architectural Interfaces and Next Compilation Strokes

With the discrete ring coordinate chain, mirror lift codec, comparator imbalance spectrum, and pullback duality closed and verified, the software specification isolates five concrete OPEN interfaces that define the next compilation cycle. The most immediate spectral target is deriving the **closed-form Walsh spectrum of the latitude in ring coordinates**: while Theorem 10.2 proves the comparator side of the pullback trade-off across all widths ($c_R(i)$ collapses onto χ_{top}), Table 2 only measures the ring-coordinate Walsh spectrum of the latitude $z(R_n[i])$ through $n = 20$ ($1{,}036{,}432$ active modes, effective participation ratio 34.6, with degree-2 holding 33% of the energy). Deriving the exact closed-form Walsh coefficients $\mathcal{W}[z \circ R_n](s)$ directly from Theorems 1, 6, and 9.3 will complete the analytical duality between value and difference readers.

In the formal languages and radial geometry container, the next stroke requires constructing a **Myhill–Nerode pairwise distinguishability proof for the radial profile language** and extending the **third-residue witness family**. The radial continuation grammar reaches 311 equivalence classes at depth 11 (≥ 312

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states) and $R_{20} = 852{,}130$ distinct profiles with a monotonically falling per-ring entropy deficit (0.0113 bits at $D = 20$). Because finite automaton growth alone cannot settle non-regularity, the newly closed reversal-ordered seam block formulas (Theorem 9.2) provide the exact boundary algebra needed to prove pairwise state distinguishability and to write the full support sets $A(q_D)$ as explicit unions over blocks. Concurrently, the two-offset center-rank witness law for $q_D = 1 \parallel 00 \parallel 1^{D-5} \parallel 00$ must be extended from $D = 3m + 2,\ m \equiv 0{,}1 \pmod 4$ to the remaining uncovered residue parameters $m \equiv 2{,}3 \pmod 4$.

Finally, in the homological and cryptographic routing containers, two synthesis bridges remain open. In CNT-05, the **all-n adder carry-seam functor** must complete the inductive proof of bounded local run-signature updates U for interior merges above $n = 6$ and construct the explicit functorial bridge mapping the carry-seam staircase simultaneously onto the Myhill–Nerode continuation state count $F(S)$ and the disjoint-square cohomology rank $b_1^{\text{disj}}(n) = \binom{n}{2} + (n-1)$. In CNT-06, **schedule-tail region pruning in wide ROBDDs** must embed the bijective 512-bit backward schedule recurrence $E: B \leftrightarrow T$ and PadValid constraints directly into the ROBDD Shannon cofactor elimination schedule, generating region-level address exclusion certificates on R_{512} that prune incompatible subtrees across multi-byte SHA-256 interfaces before intermediate decision-node counts scale exponentially.

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